

2-Phase Interleaved Sync-Buck Converter

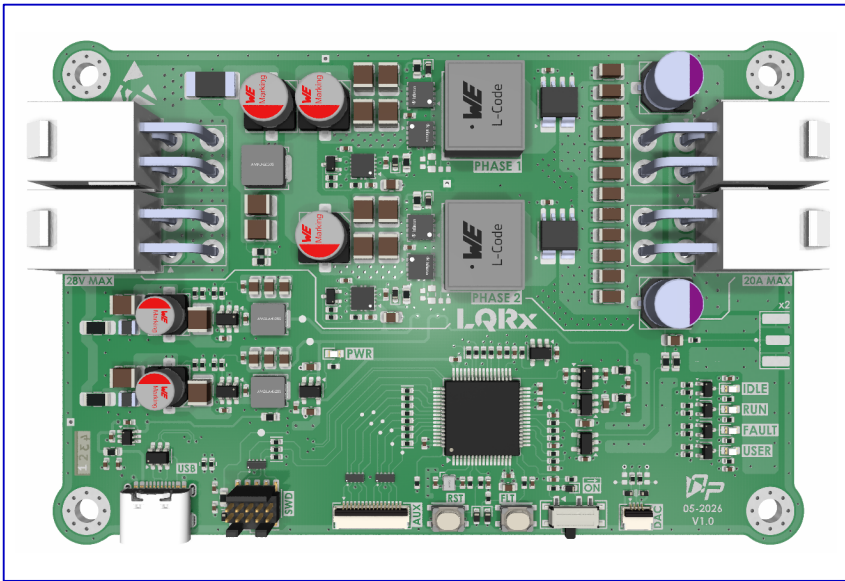
Digitally Controlled Power Conversion with STM32G474

Revision 1.0

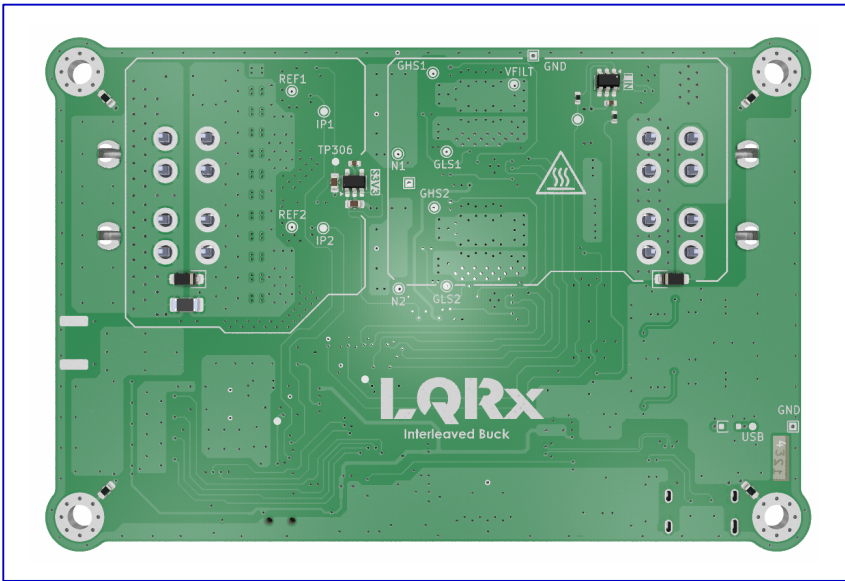
Table of Contents

- A Project Overview
- B Block Diagram
- C Top Hierarchical View
 - 1 Power State
 - 2 Power Connectors
 - 3 Power Supplies
 - 4 Microcontroller Power
 - 5 Microcontroller IOs
 - 6 Analog Front-End
 - 7 Interface Interconnects
 - 8 User LEDs
 - 9 User Interface
- D Power Supply Sequencing
- E Revision History

TOP VIEW



BOTTOM VIEW



Legend

DESIGN NOTE:
Text for information design.

DEBUG NOTE:
Text for Debug notes.

DESIGN NOTE:
Text for design guideline

DESIGN NOTE:
Text for critical design

LAYOUT NOTE:
Text for layout guideline

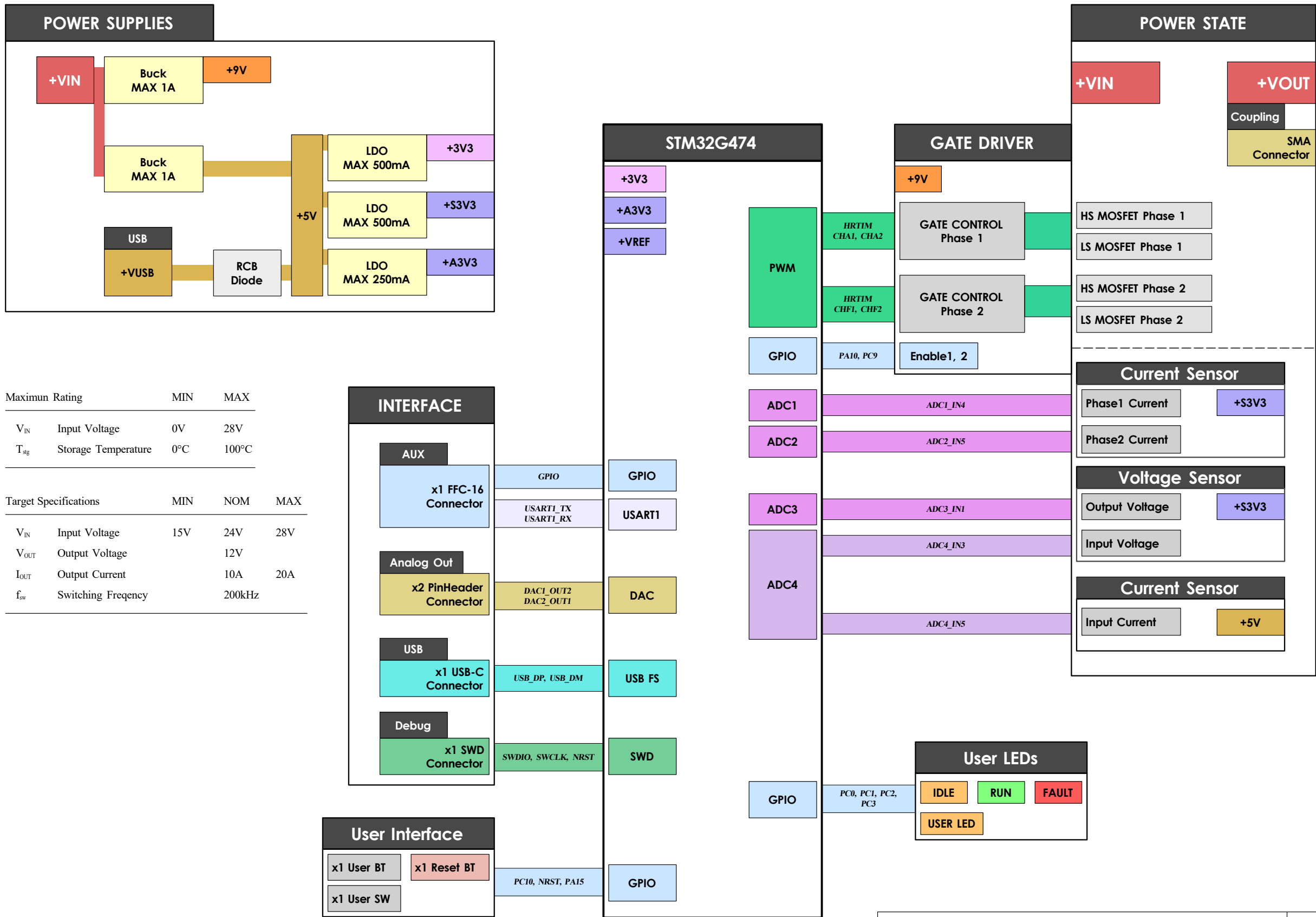
LAYOUT NOTE:
Text for Critical layout

General Note
Warning Note
Text to be added to silkscreen.

LQRx

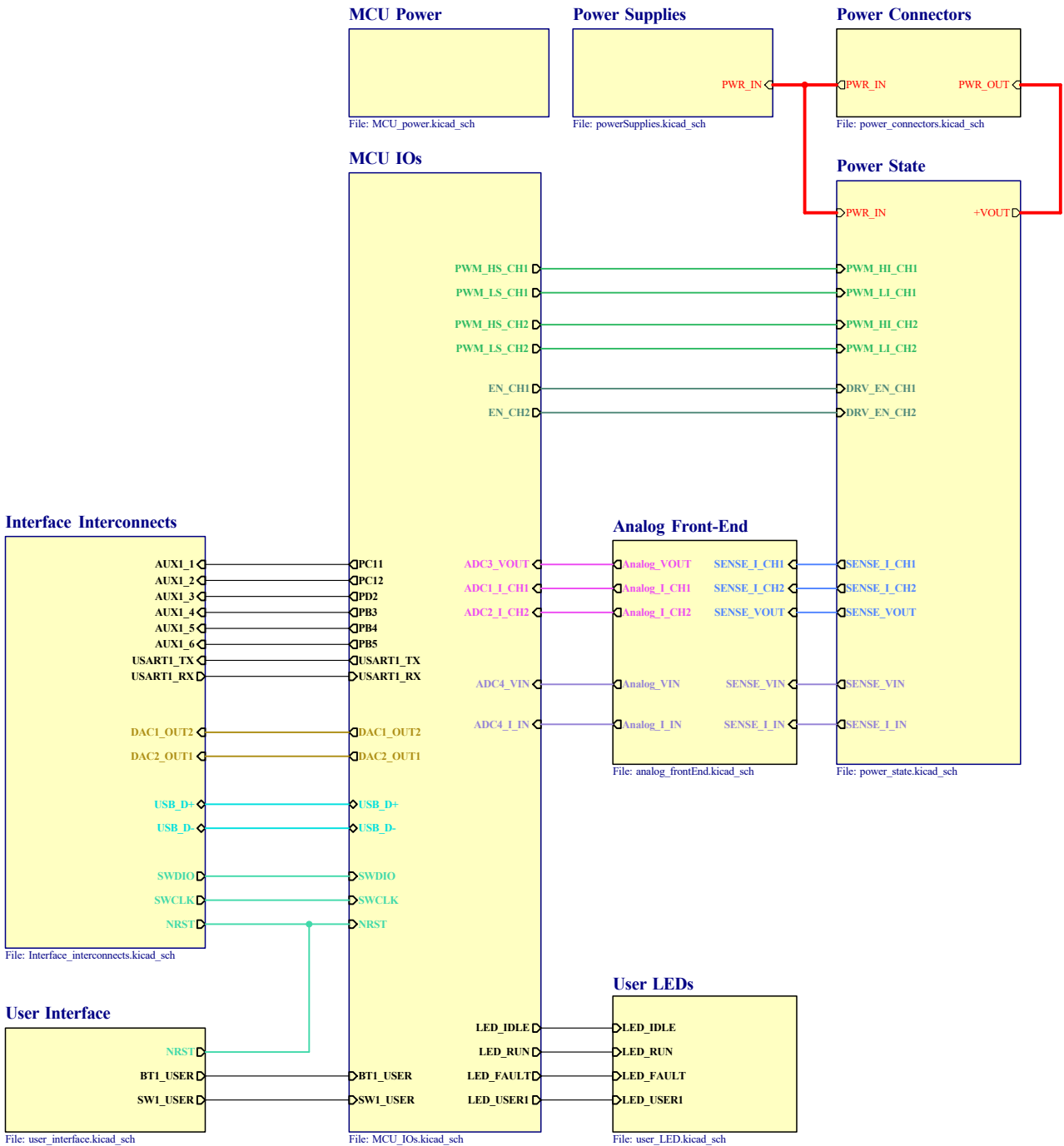
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Project Overview Cover Page	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.0 Date: 28/05/2026
	Sheet: /	Size A3	Sheet: A of 9
KiCad E.D.A. 10.0.3		File: 2Phase_SynchBuck.kicad_sch	

[B] Block Diagram



Project: 2-Phase Interleaved Sync-Buck Converter			LQRx		
Title: Block Diagram (Top Level View)	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:		
	Reviewed by:	Date:	Rev: 1.0	Date: 28/05/2026	
	Sheet: /Block Diagram/			Size	Sheet:
	KiCad E.D.A. 10.0.3	File: Block_diagram.kicad_sch		A3	B of 9

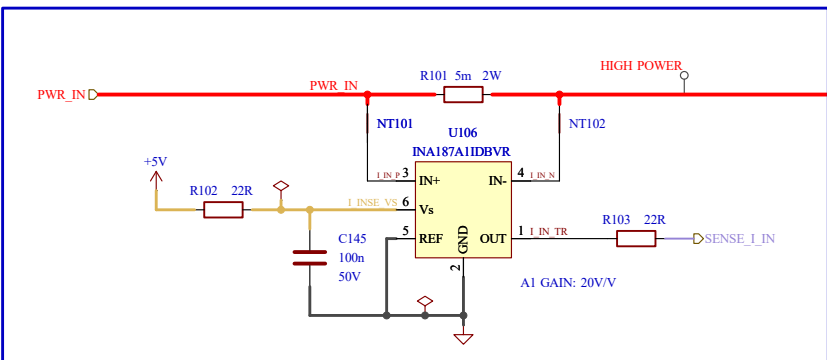
[C] Top Hierarchical View



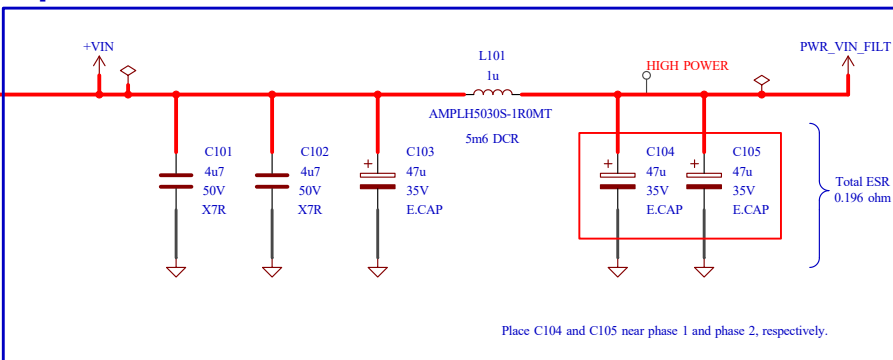
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx		
Title: Top Hierarchical View (Top Level View)	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:	
	Reviewed by:	Date:	Rev: 1.0	Date: 28/05/2026
	Sheet: /Top/		Size A3	Sheet: C of 9
KiCad E.D.A. 10.0.3		File: Top_hierarchical.kicad_sch		

[1] Power State

I_{IN} Sense

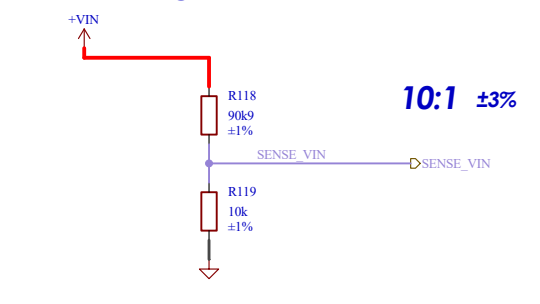


Input Filter

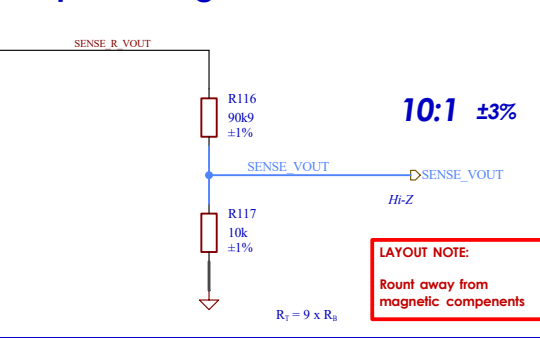


DESIGN NOTE:
GRT32ER71H475KE01
Input Capacitors 4.7uF x8 Derating -33% @24V = 25.2uF
 $C_d = 4^\circ C = 100\mu F$
 $Z_s = \sqrt{L/C} = \sqrt{1/25.2} = 0.199\Omega$
 $R_d = Z_s^2 / (Z_s/Q - DCR)$
 $Q=0.9, DCR=15m \Rightarrow R_d = 0.192\Omega$
<Target 100uF ESR 0.192ohm>
Used Electrolytic Cap.(47uF, ESR 0.39ohm) x2 for stability
 $Z_{filter} Peak -13.78dB @28kHz$
 $<Atten. -20.5dB @ 200kHz$

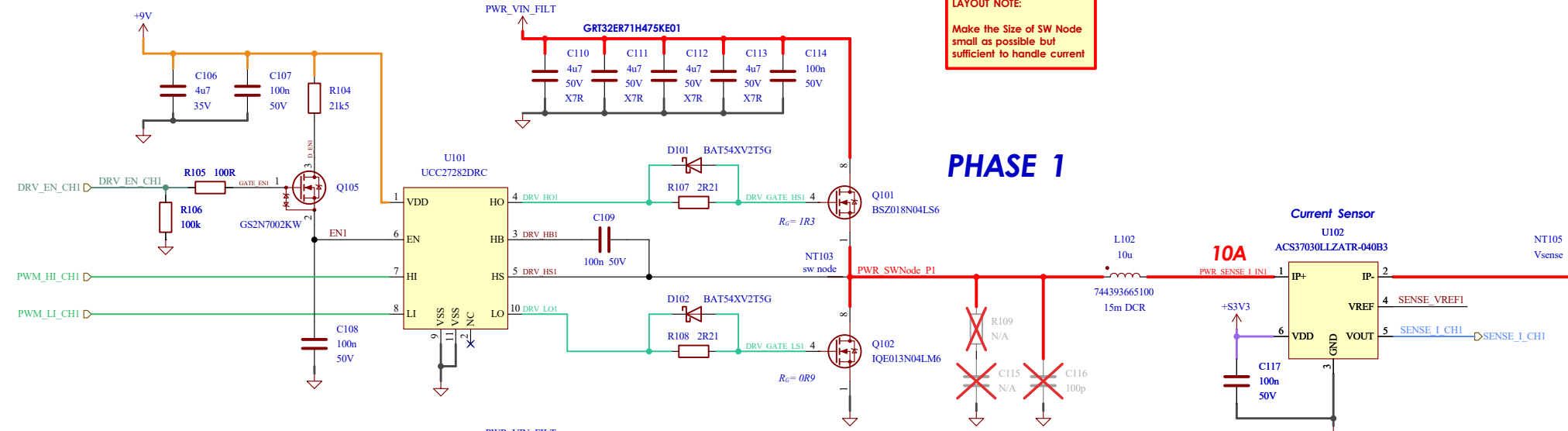
Input Voltage Sense



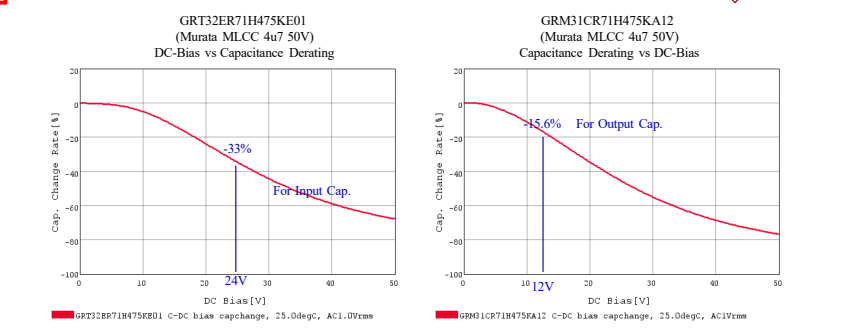
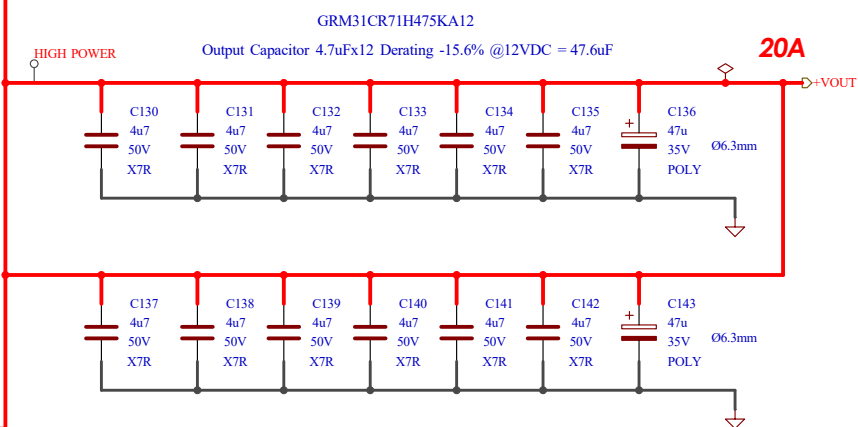
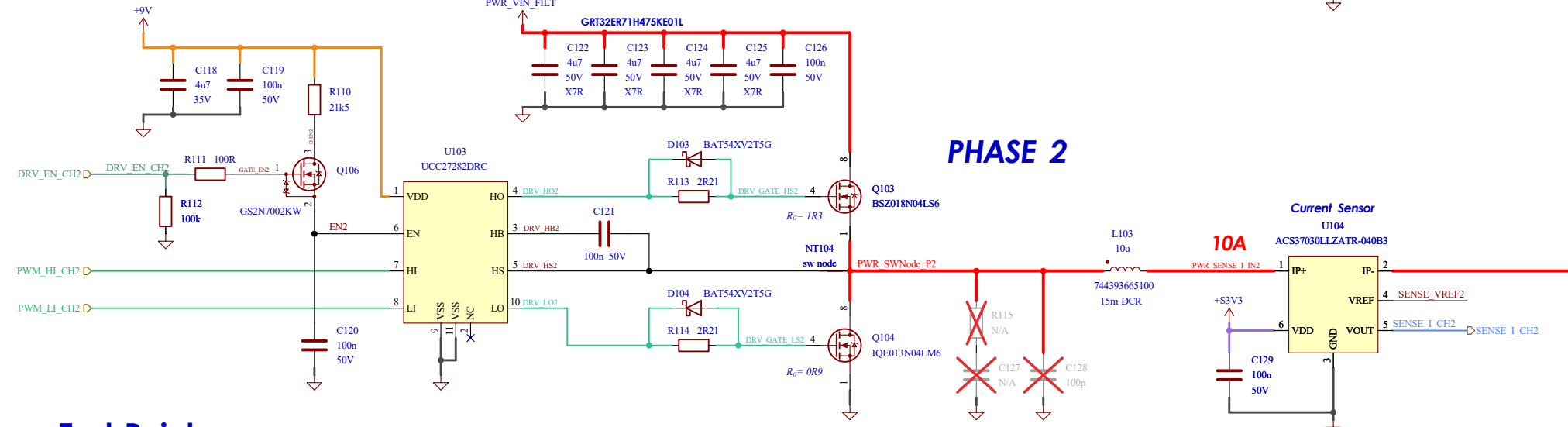
Output Voltage Sense



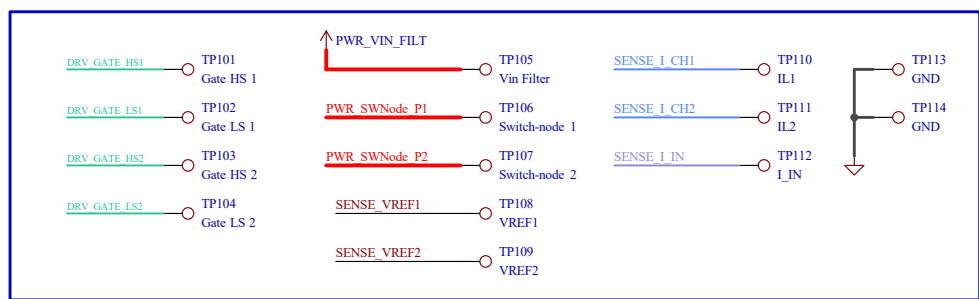
PHASE 1



PHASE 2



Test Point



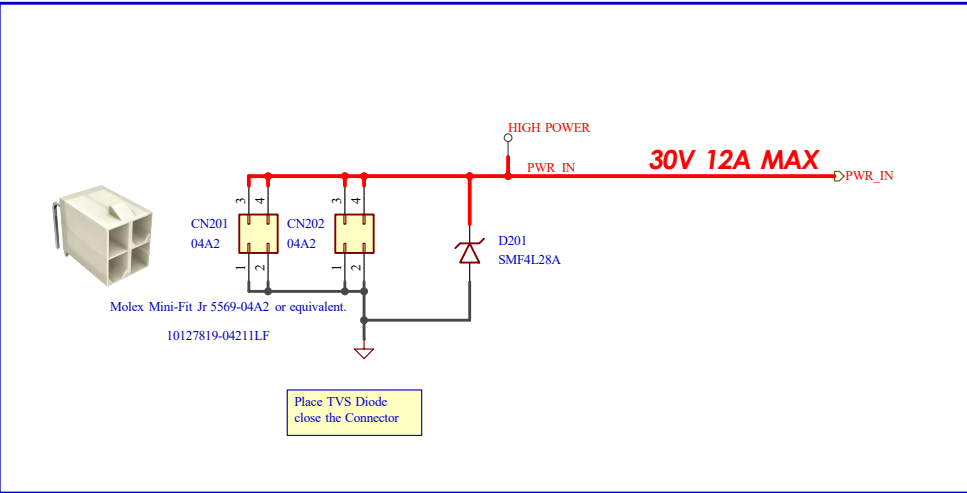
DEBUG NOTE:
If Switch-node are ringing
C116 and C128 are capacitor to find R C values in the snubber circuit.
 $C_{ps} = C_{sn} [f_s^2 / (f_s^2 - f_r^2)]$
 $L_{ps} = 1 / [2\pi f_s C_{ps}]$
 $C_{ms} = (4-8) C_{ps}$
 $R_{ms} = 1/Q [\sqrt{L_{ps} / C_{ms}}]$



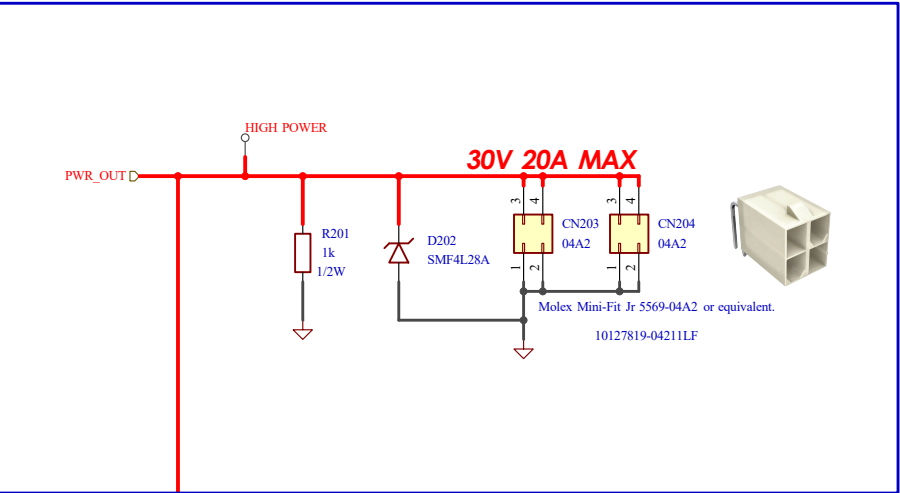
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Power State 2-Phase Buck & Gate Driver	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.0 Date: 28/05/2026
Sheet: /Top/Power State/ File: power_state.kicad_sch		Size: A3	Sheet: 1 of 9
KiCad E.D.A. 10.0.3			

[2] Power Connectors

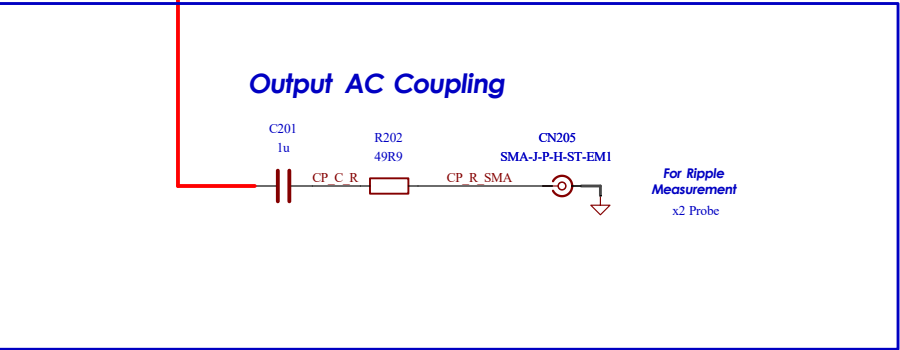
Input Connector



Output Connector



Output AC Coupling

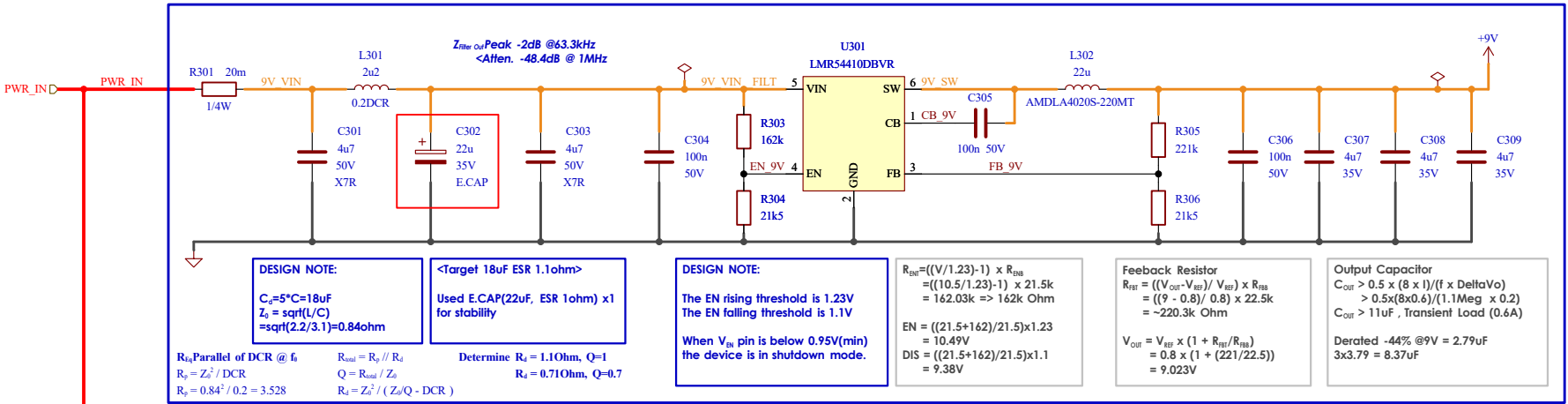


Project: 2-Phase Interleaved Sync-Buck Converter		LQRx		
Title: Power Connectors	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:	
	Reviewed by:	Date:	Rev: 1.0	Date: 28/05/2026
	Sheet: /Top/Power Connectors/		Size A3	Sheet: 2 of 9
KiCad E.D.A. 10.0.3		File: power_connectors.kicad_sch		

[3] Power Supplies

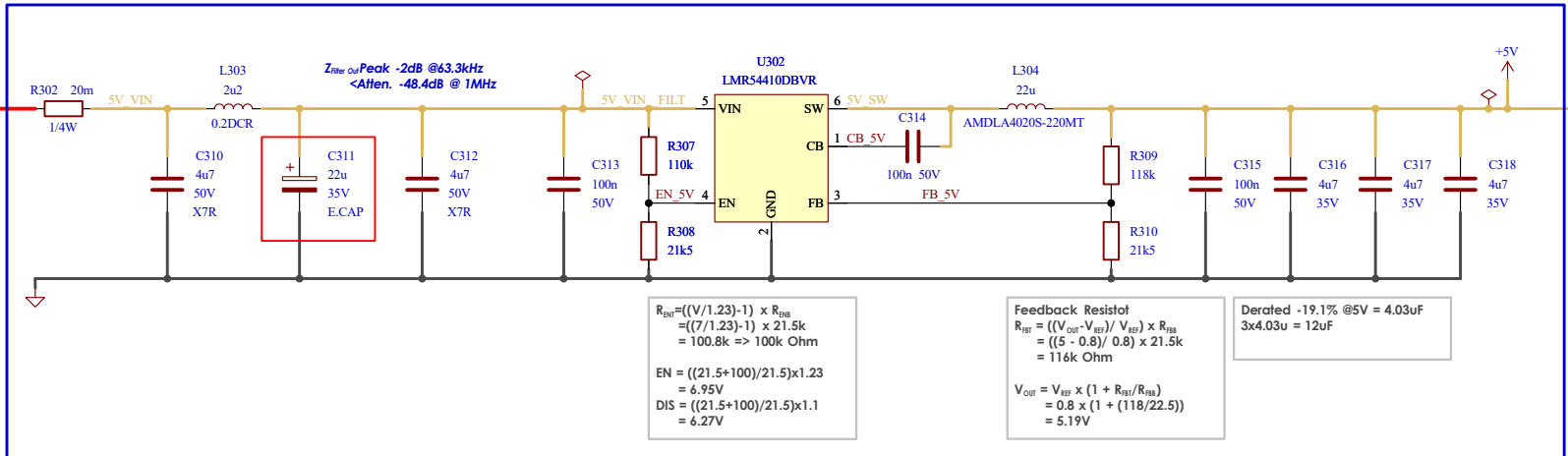
Buck Converter 9V 1A (1.1MHz)

*Enable when V_{IN} is above 11V

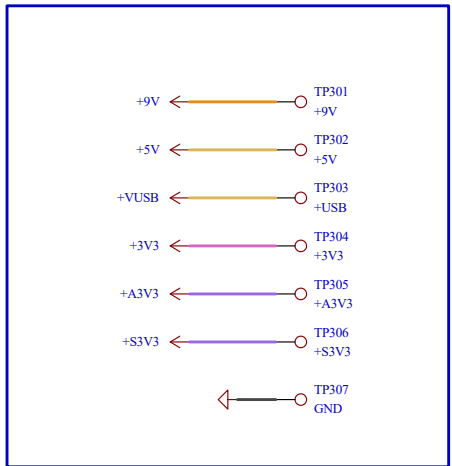


Buck Converter 5V 1A (1.1MHz)

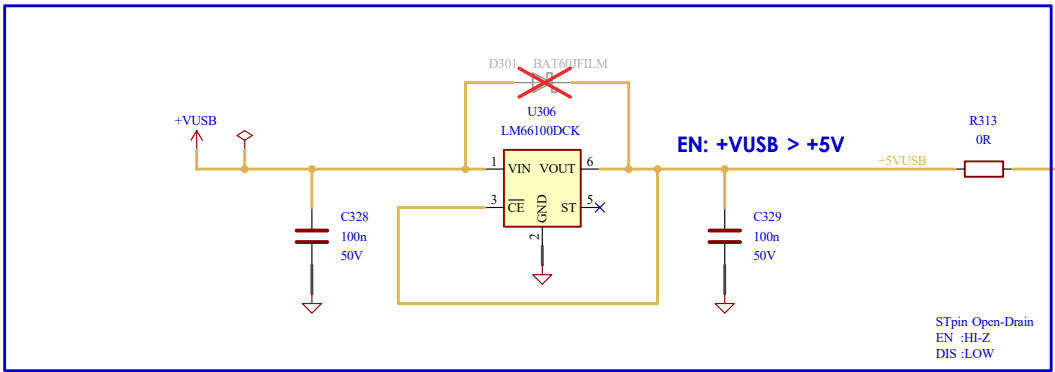
*Enable when V_{IN} is above 7V



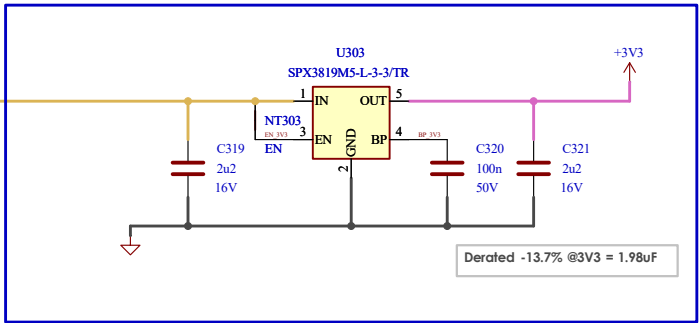
Test Point



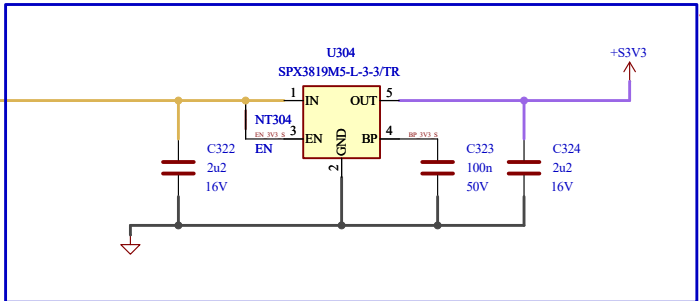
Reverse Current Blocking



LDO 3V3 500mA (Digital path)

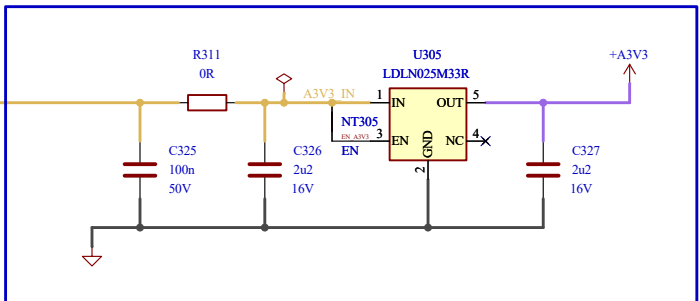


LDO 3V3 500mA (Sensor path)

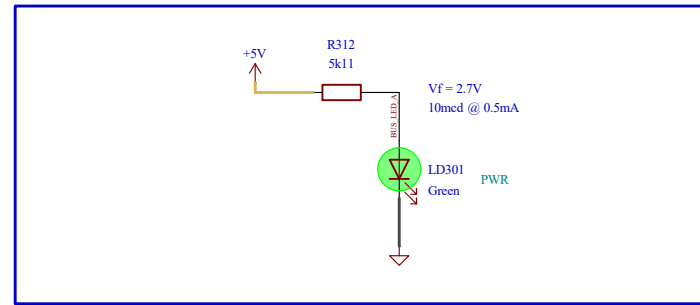


LAYOUT NOTE:
Place this power supply near the current sensor
On Page 1

LDO 3V3 250mA (Analog path)



LED Indicators



Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Power Supplies Auxiliary 9V 5V 3V3	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.0
KiCad E.D.A. 10.0.3	Sheet: /Top/Power Supplies/ File: powerSupplies.kicad_sch		Date: 28/05/2026
	Size A3		Sheet: 3 of 9

[illegible]

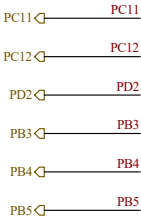
Project: 2-Phase Interleaved Sync-Buck Converter				LQRx			
Title: Microcontroller Power STM32G474Rx		Drawing by: NATCHPAI		Date: 08/04/2026		Variant:	
		Reviewed by:		Date:		Rev: 1.0	Date: 28/05/2026
		Sheet: /Top/MCU Power/				Size A3	Sheet: 4 of 9
KiCad E.D.A. 10.0.3		File: MCU_power.kicad_sch					
6		7		8			

[5] Microcontroller IOs

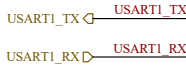
Interconnects

Auxiliary Port 1

GPIO



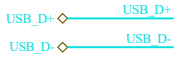
USART



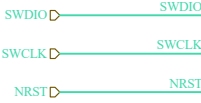
Analog Output (DAC)



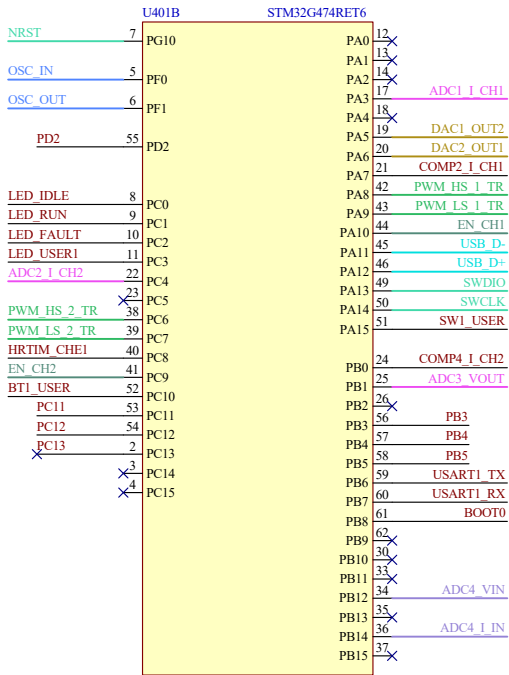
USB2.0 FS



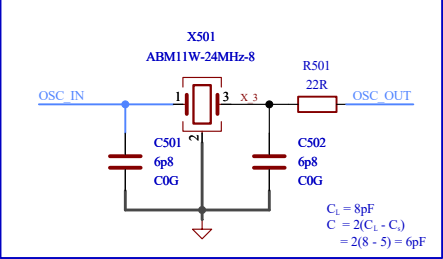
Serial Wire Debug



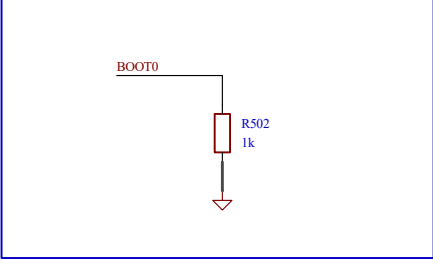
STM32G474 I/O



HSE Clock (24MHz)



BOOT0



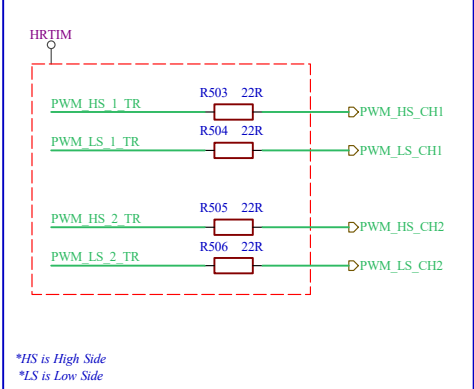
User Interface



Test Point



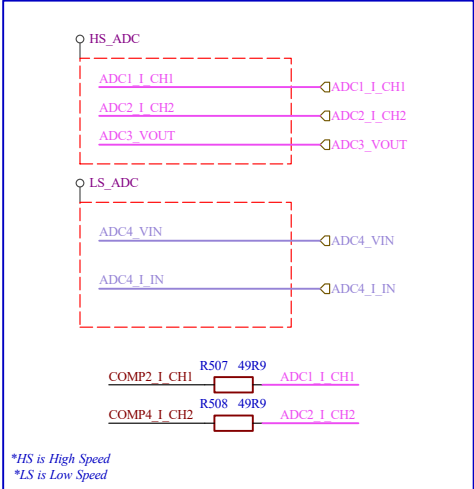
High Resolution Timer



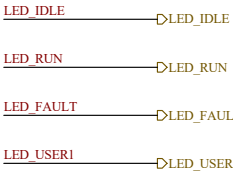
Enable Gate Driver



ADC



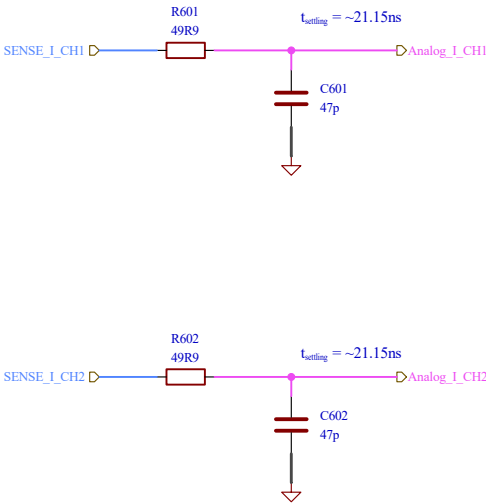
LEDs



Project: 2-Phase Interleaved Sync-Buck Converter				LQRx				
Title: Microcontroller IOs STM32G474Rx			Drawing by: NATCHPAI		Date: 08/04/2026		Variant:	
			Reviewed by:		Date:		Rev: 1.0	Date: 28/05/2026
			Sheet: /Top/MCU IOs/ File: MCU_IOs.kicad_sch				Size A3	Sheet: 5 of 9
KiCad E.D.A. 10.0.3								
6			7		8			

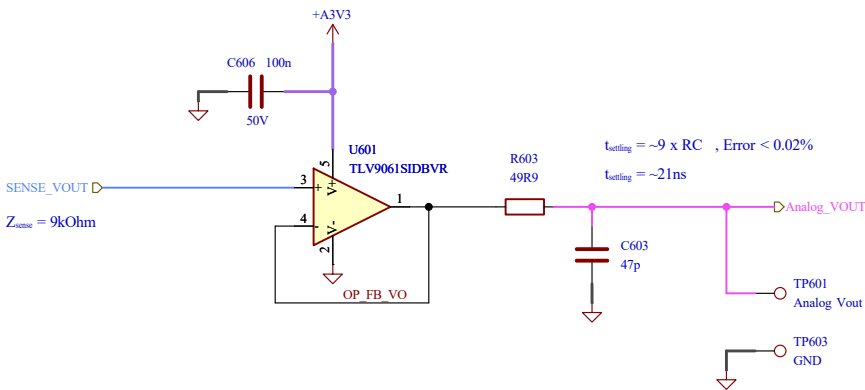
[6] Analog Front-End

Current Path (High Speed)



ADC1, 2 (48.571429MHz) 12-Bit Res.
 $t_{ACQ} = 2.5 \text{ Cycle} = 51.47\text{ns}$
 $R_{AIN(MAX)} = 100R$ $C_{ADC} = 10\text{pF}$

Output Voltage Path (High Speed)



LAYOUT NOTE:

Placing OpAmp and RC Filter close to the ADC input or Analog Zone

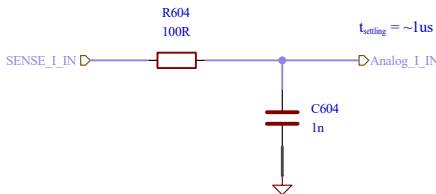
LAYOUT NOTE:

Round away from magnetic components

t_{ACQ} must greater than $t_{settling}$ for accuracy.

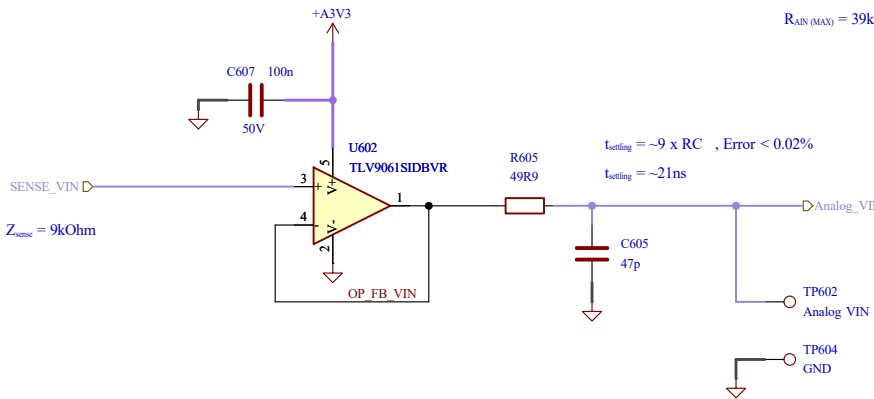
ADC3 (42.5MHz) 12-Bit Res.
 $t_{ACQ} = 2.5 \text{ Cycle} = 58.8\text{ns}$
 $R_{AIN(MAX)} = 100R$ $C_{ADC} = 10\text{pF}$

Input Current Path (Low Speed)



ADC4 (42.5MHz) 12-Bit Res.
 $t_{ACQ} = 640.5 \text{ Cycle} = 15.07\mu\text{s}$
 $R_{AIN(MAX)} = 39k$ $C_{ADC} = 10\text{pF}$

Input Voltage Path (Low Speed)



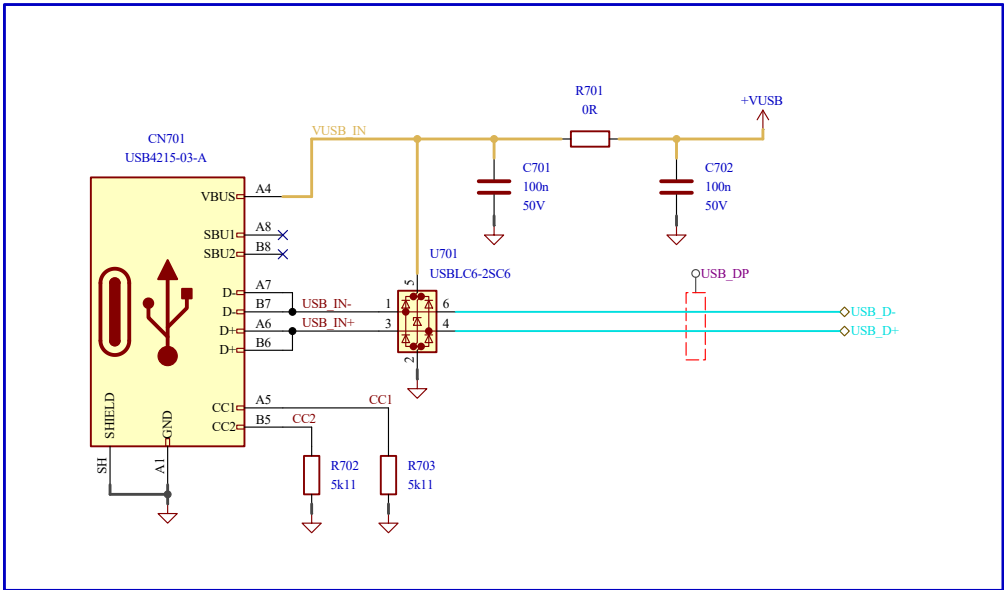
No buffer and short this JP: $t_{settling} = \sim 4\mu\text{s}$

ADC4 (42.5MHz) 12-Bit Res.
 $t_{ACQ} = 640.5 \text{ Cycle} = 15.07\mu\text{s}$
 $R_{AIN(MAX)} = 39k$ $C_{ADC} = 10\text{pF}$

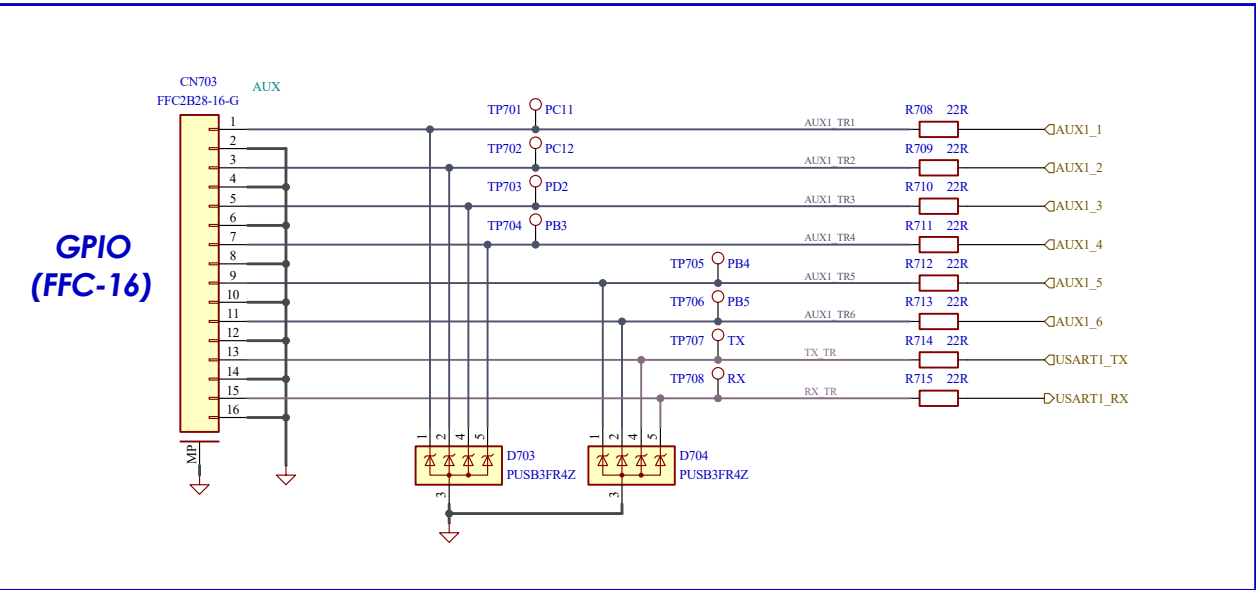
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx		
Title: Analog Front-End	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:	
	Reviewed by:	Date:	Rev: 1.0	Date: 28/05/2026
	Sheet: /Top/Analog Front-End/		Size A3	Sheet: 6 of 9
KiCad E.D.A. 10.0.3		File: analog_frontEnd.kicad_sch		

[7] Interface Interconnects

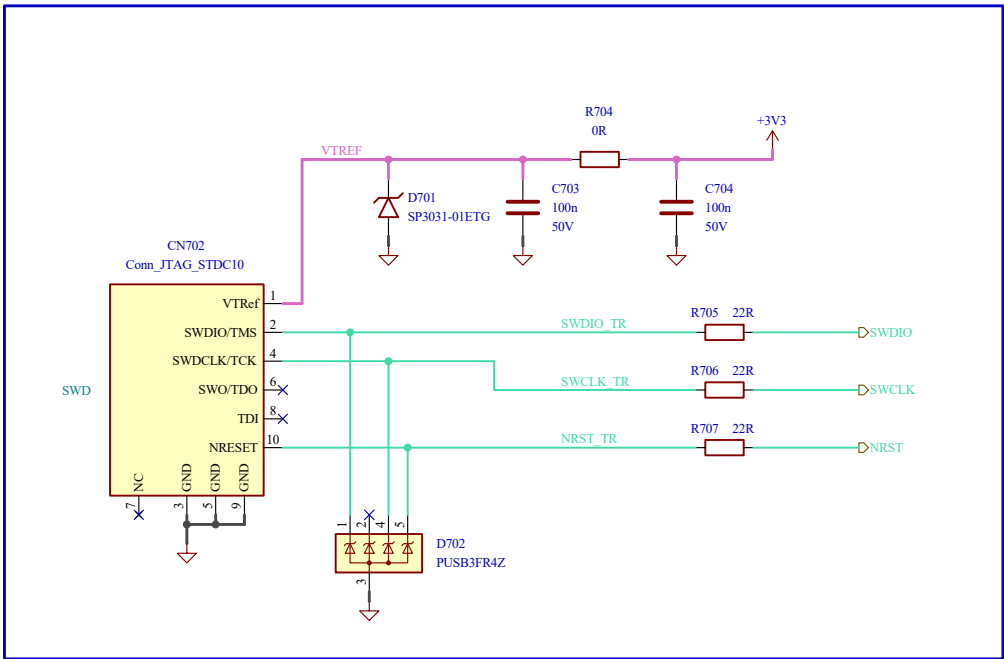
USB-C Connector



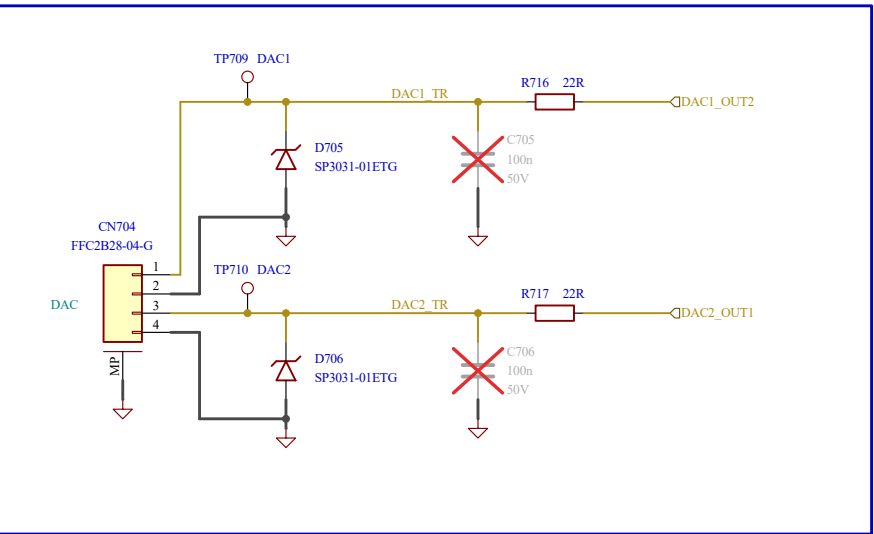
AUX Connector



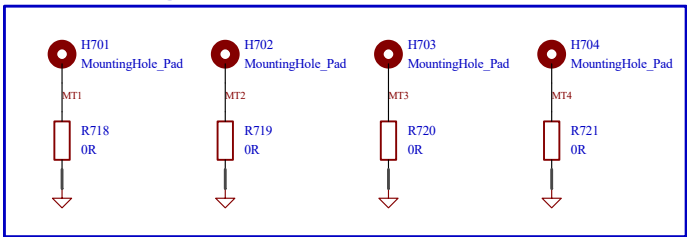
SWD Connector



Analog Output



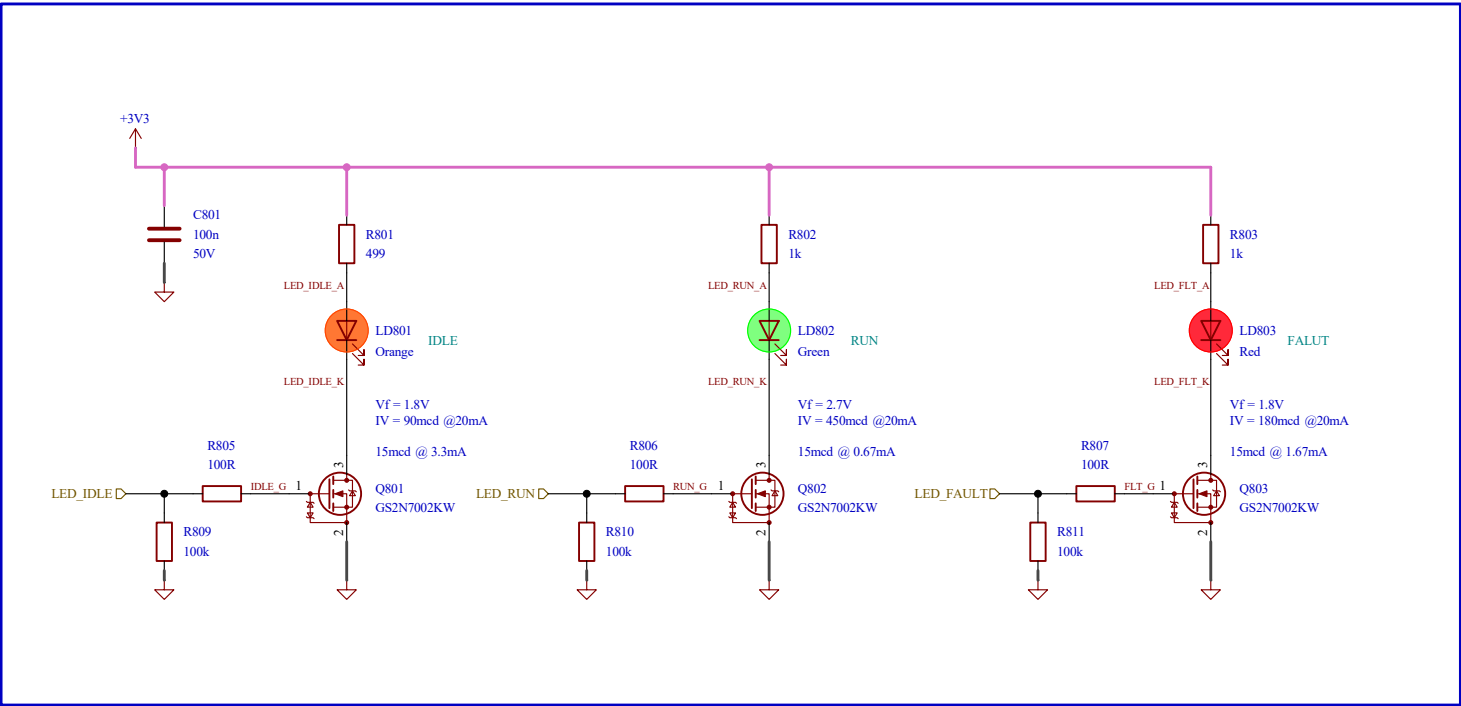
Mounting Hole



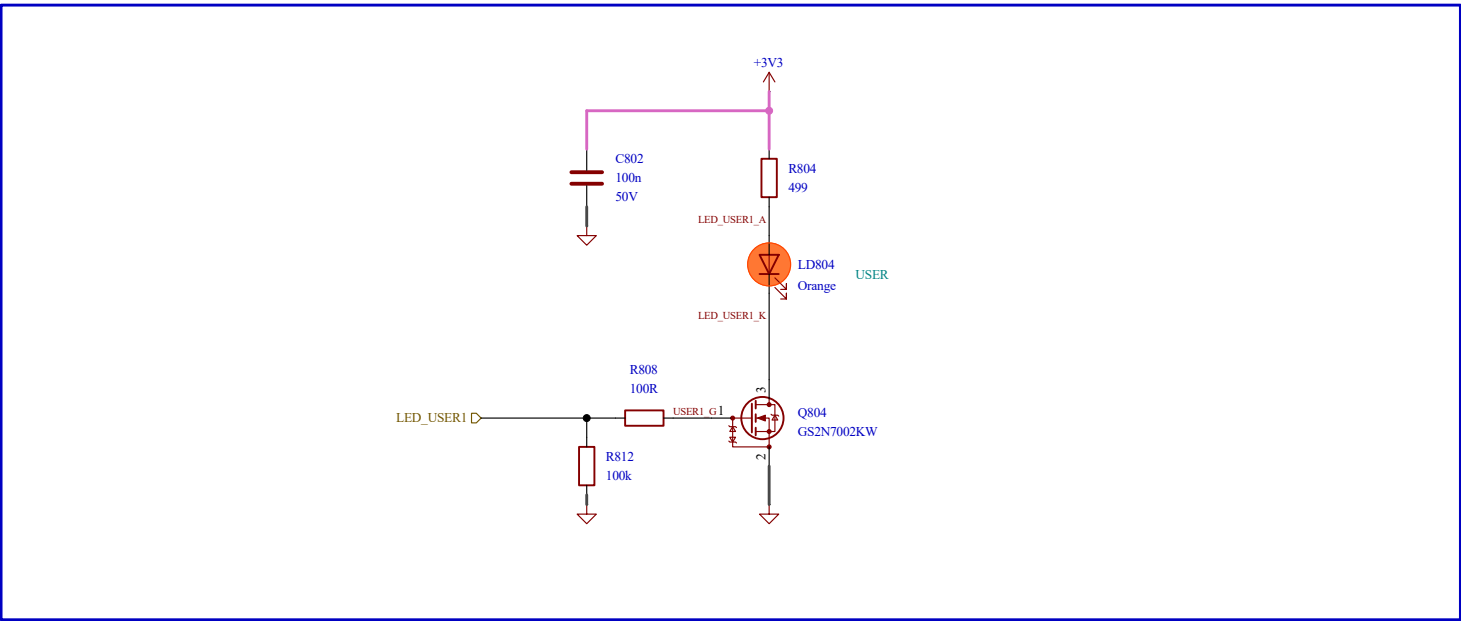
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Interface Interconnects	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.0
KiCad E.D.A. 10.0.3	Sheet: /Top/Interface Interconnects/		Size A3
	File: Interface_interconnects.kicad_sch		Sheet: 7 of 9

[8] User LEDs

Status LEDs



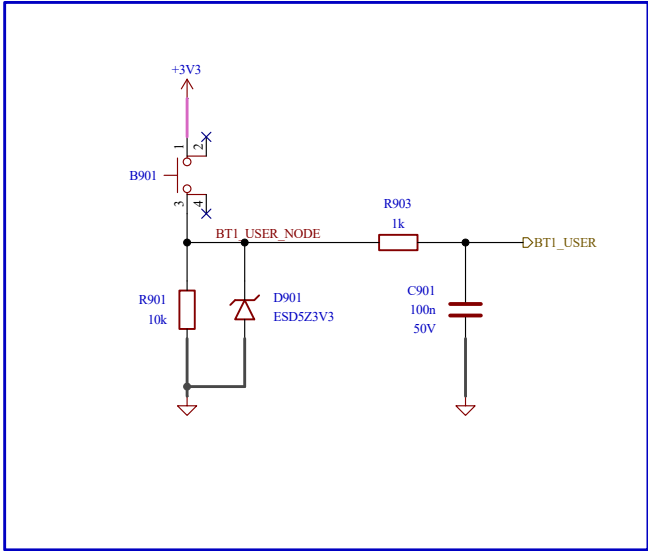
User LEDs



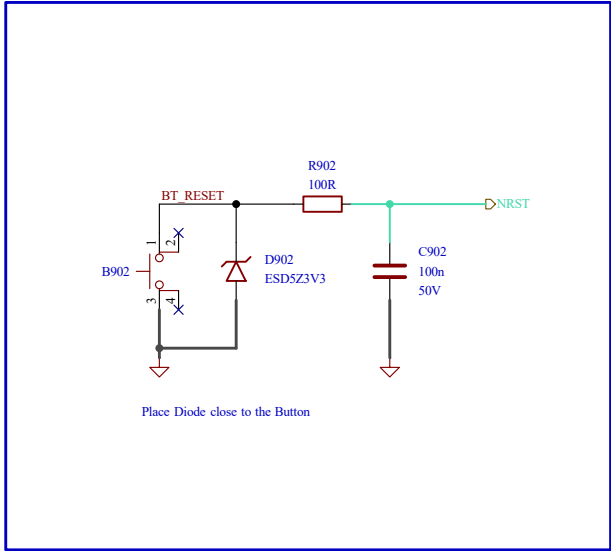
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx		
Title: User LEDs Status LEDs	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:	
	Reviewed by:	Date:	Rev: 1.0	Date: 28/05/2026
	Sheet: /Top/User LEDs/		Size A3	Sheet: 8 of 9
KiCad E.D.A. 10.0.3		File: user_LED.kicad_sch		

[9] User Interface

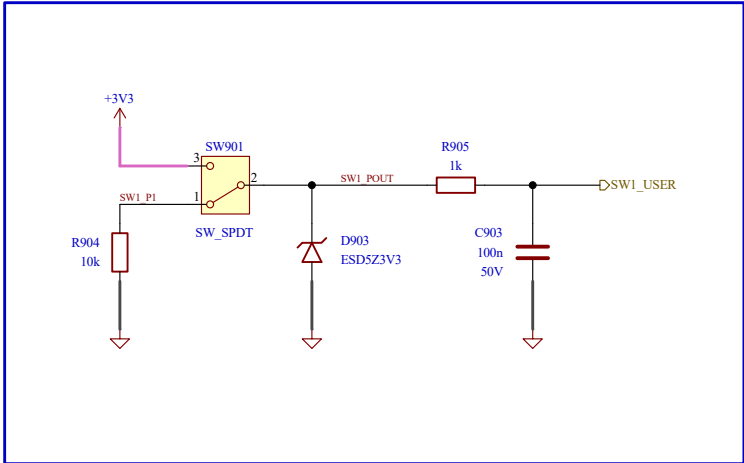
User Button



Reset Button

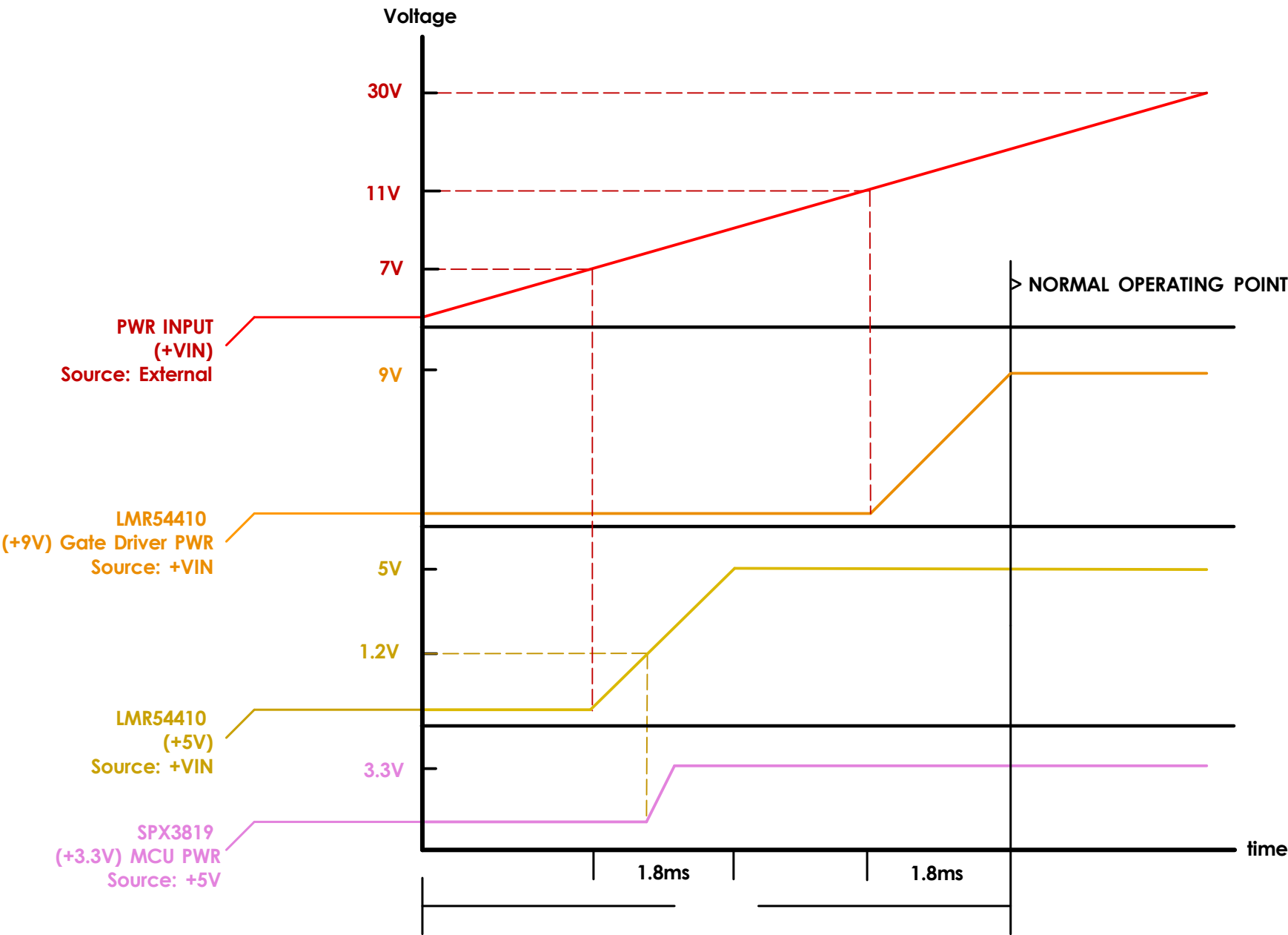


User Switch



Project: 2-Phase Interleaved Sync-Buck Converter		LQRx		
Title: User Interface Button & Switch	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:	
	Reviewed by:	Date:	Rev: 1.0	Date: 28/05/2026
	Sheet: /Top/User Interface/		Size A3	Sheet: 9 of 9
KiCad E.D.A. 10.0.3		File: user_interface.kicad_sch		

[D] Power Supply Sequencing



Project: 2-Phase Interleaved Sync-Buck Converter		LQRx		
Title: Power Supply Sequencing	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:	
	Reviewed by:	Date:	Rev: 1.0	Date: 28/05/2026
	Sheet: /Power Supply Sequencing/		Size A3	Sheet: D of 9
KiCad E.D.A. 10.0.3		File: pwr_sequencing.kicad_sch		

[E] Revision History

Revision 1.018/05/2026

- First Version of Schematic (Printed)
- Added 2x(47uF Polymer Cap.) on Output State
- Added Rsense on Input Terminal in Power Supplies Circuits
- Added Input Current Sensor on Power State
- Added Turn-Off Diode in parallel with R-Gate in Gate Driver Circuits
- Added Series Termination resistor on HRTIM's Line
- Changed Input and Output Connectors from 6 PIN to 2x(4 PIN)
- Changed GPIO's PinHeader to FFC-16 P0.5 Connector
- Changed DAC's PinHeader to FFC-4 P0.5 Connector
- Removed Temp. Sensor

Project: 2-Phase Interleaved Sync-Buck Converter				LQRx		
Title: Revision History	Drawing by: NATCHPAI		Date: 08/04/2026		Variant:	
	Reviewed by:		Date:		Rev: 1.0	Date: 28/05/2026
	Sheet: /Revision History/				Size A3	Sheet: E of 9
	KiCad E.D.A. 10.0.3				File: RevHistory.kicad_sch	
6		7		8		F